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Introduction

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Super-resolution structured illumination microscopy: past, present and future

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Structured illumination microscopy (SIM) has emerged as an essential technique for three-dimensional (3D) and live-cell super-resolution imaging. However, to date, there has not been a dedicated workshop or journal issue covering the various aspects of SIM, from bespoke hardware and software development and the use of commercial instruments to biological applications. This special issue aims to recap recent developments as well as outline future trends. In addition to SIM, we cover related topics such as complementary super-resolution microscopy techniques, computational imaging, visualization and image processing methods.

This article is part of the Theo Murphy meeting issue 'Super-resolution structured illumination microscopy (part 1)'.

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1. Introduction

Fluorescence light microscopy is a core technique in life sciences that has contributed to countless major discoveries. However, the diffraction limit, first described by Ernst Abbe in 1873 [1], has restricted the optical resolution to about half the wavelength of the light used, i.e. 200–300 nm in the lateral directions (x and y), and to about the wavelength of light, i.e. 500–800 nm along the optical axis (z). In the past two decades, several super-resolution microscopy (SRM) approaches have been developed that overcome this barrier and push the spatial resolution to the 10–150 nm range, thus closing the gap to electron microscopy [2–5]. These SRM techniques can be further divided into single-molecule localization microscopy (SMLM) [6] that includes techniques like photoactivatable localization microscopy (PALM) [7,8] and stochastic optical reconstruction microscopy (STORM) [9], stimulated emission depletion (STED) [10–12] microscopy, and structured illumination microscopy (SIM) [13,14]. In recognition of these breakthrough developments Eric Betzig, Stefan Hell and William E. Moerner were awarded the Nobel prize for Chemistry in 2014 [15].

Recent SRM developments have focused on combining localization-based microscopy with modulated illumination to push the resolution (precision) towards a few nanometres as in MINFLUX and SIMFLUX [16–19]. However, an increase in resolution requires an increase in local light dosage, which increases photobleaching/phototoxicity (SIM, STED, SMLM) or requires the need to introduce spatial and temporal sparsity (MINFLUX) making imaging comparably slow. Moreover, most SRM techniques mentioned above rely primarily on chemically fixed (i.e. dead) cells, are restricted in the imaged sample volume (two dimensions, single plane, small field-of-views) or are restricted to imaging sparse single entities such as vesicles or single molecules.

Super-resolution linear SIM is a notable exception [20]. By making use of frequency mixing when exciting samples with a patterned illumination followed by computational unmixing and reconstruction, SIM achieves a twofold resolution increase over conventional diffraction-limited fluorescence microscopy in two dimensions or three dimensions (figure 1). While the numerical resolution improvement is moderate compared to other SRM techniques, it pushes SIM into an application sweet spot with many macromolecular structures and their range of motion falling in the size range of 100–300 nm [21–23]. By not asking for the highest spatial resolution, SIM is less demanding in terms of photon budget and therefore more compatible with live-cell imaging [24–29]. In addition, it offers high contrast volumetric imaging with relatively large field-of-views and comparably high temporal resolution which makes it suitable for high-content and live-cell imaging [30–32]. Taken together, the uniquely balanced combination of properties has made SIM remarkably successful for a wide range of biological applications promoting new discoveries [22, 33–48].

2. Recent and future developments in SIM

SIM has been available in commercial instruments for more than 10 years and has been successfully established in many laboratories and core facilities around the world. However, wider dissemination of SIM has been curtailed by the complexity of instruments and its proneness to reconstruction artefacts when sample properties, system calibration and parameter settings are not carefully matched [49]. Thus, to lower the activation energy for research labs to venture into super-resolution SIM, many recent developments aim at further performance/resolution enhancement and 'democratizing' SIM by

— using nonlinear SIM approaches [50–53] or by combinations with single-molecule imaging [16,17,54–56],
— exploiting correlative and combinatorial fluorescence imaging approaches with multifocal [57,58], 2-photon [59–61] and light-sheet microscopy [62,63],

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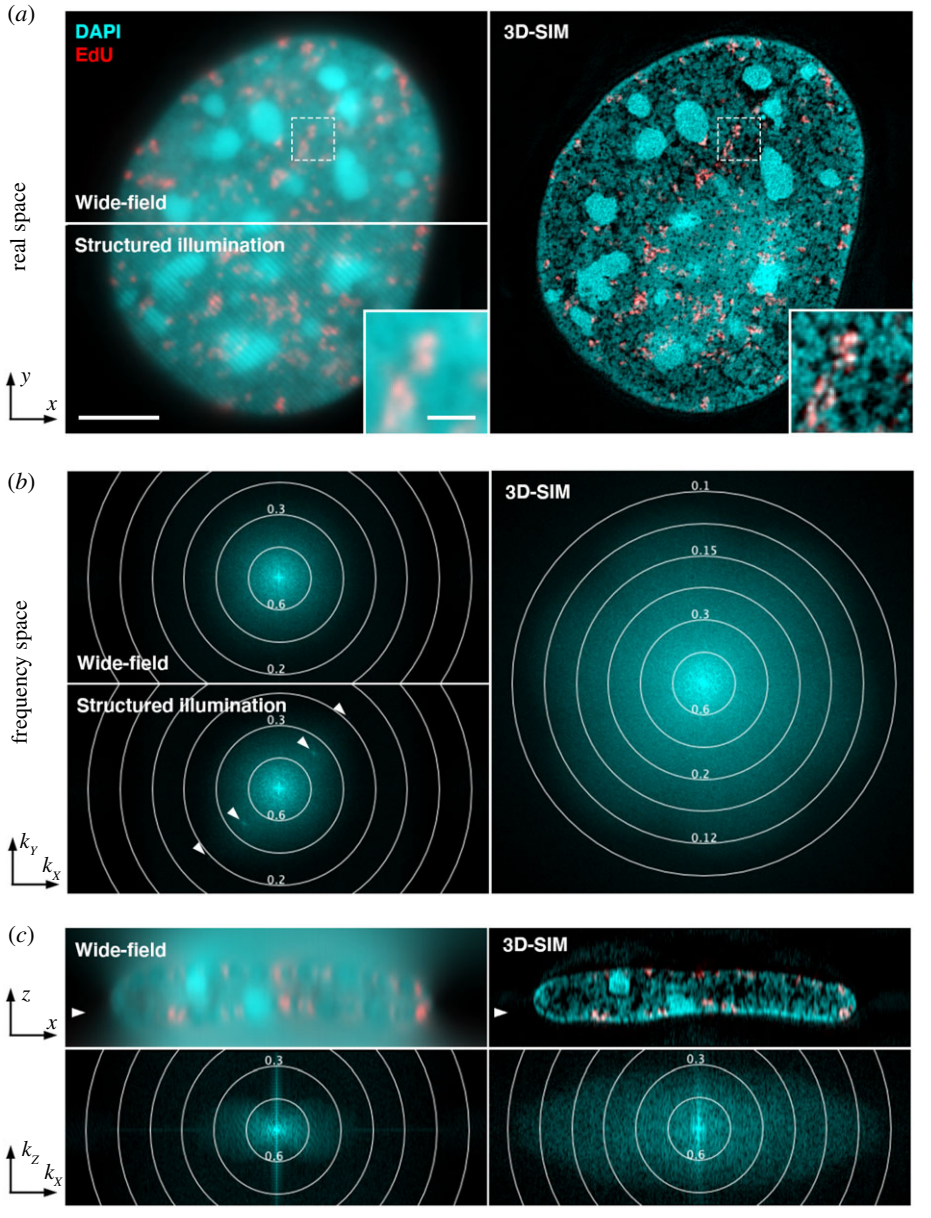


Figure 1. Biological super-resolution imaging with 3D-SIM. (*a*) Mouse C127 mammary epithelial cell nucleus replication labelled with 5-ethenyl-2′-deoxyuridine (EdU, red) for 15 min before fixation with formaldehyde. The thymidine analogue EdU is incorporated into newly synthesized DNA of S-phase cells (here mid-to-late S phase) and detected via click-chemistry with Alexa Fluor 594 azide. DNA is labelled with 4′, 6-diamidino-2-phenylindole (DAPI, cyan). Single *z*-section of an image stack is shown with conventional wide-field illumination (top, left), structured illumination (1 of 15 raw images acquired per *z*-plane with laterally shifted and rotated stripes; bottom, left), and after 3D-SIM reconstruction (right). Note that 3D-SIM resolves higher-order domain organization of chromatin and DNA-free interchromatin regions (inset), as well as the location of nuclear pores in the peripheral chromatin layer visible as DAPI void dots in the central region of the nucleus. Scale bar: 5 and 1 µm (inset). (*b*) Corresponding frequency distribution of the DAPI signal in Fourier (reciprocal) space. Concentric rings indicate the respective spatial resolution in µm. Spots in the raw SI frequency plot (arrowheads) correspond to first- and second-order stripes in the image (generated by a three-beam interference; only coarse first-order stripes are visible in the image). (*c*) Orthogonal cross-section and corresponding frequency distribution of the same dataset. The arrowheads indicate the position of the *z*-section shown in (*a*). Note the twofold extended frequency distribution in the reconstructed data in both lateral and axial direction, that includes the filling-in of the missing frequencies along the *z*-axis of the wide-field frequency plot. (Online version in colour.)

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— implementing the technique into smaller and more cost-efficient set-ups [64–66],
— making the method more robust against artefacts using alternative illumination schemes [67–69] and/or intelligent data processing [70,71],
— increasing its application range, e.g. by the implementation of adaptive optics [72–74], and cryo-imaging [47,75].

Important advancements are being made to improve and simplify the instrumentation. For instance, newly designed optical set-ups using an open-sources toolbox [65] or waveguide-based photonic chips [70], and open-source software development [76] will reduce the cost of SR imaging by enabling SIM on smaller and more affordable devices. Moreover, next-generation sCMOS cameras offer increased frame rates of up to 500 fps and higher sensitivity with reduced noise levels to enable faster high-quality SIM imaging for observing cellular dynamics [77–79]. For maximal imaging speed, rolling shutter schemes can be combined with SIM data acquisition [28,80].

In parallel, the rapidly evolving field of computational microscopy is paving the way to circumvent the resolution limit without the necessity of specialized hardware reducing the complexity of instrumentation. For example, machine learning (ML) assisted denoising has been demonstrated to improve SIM reconstructions by reducing artifacts and boosting the effective resolution for 2D-SIM [65,66,70,71,81–84]. Thus, by combining physical data acquisition with smart image processing algorithms, previously inaccessible information can be recovered. At the same time, this brings up new problems of validation of the results from ML methods, therefore calling for new standards on the performance of these algorithms. Finally, adaptive optics (AO) implementations have demonstrated great potential to correct sample aberrations and restore performance when imaging challenging samples [72–74]. However, AO does not solve the problem of reduced illumination pattern contrast in the presence of significant out-of-focus background. Thus, combinations of AO-SIM with light sheet illumination, 2-photon or photoswitching methods [57–63] present avenues for next developments.

Frequently asked questions in super-resolution structured illumination microscopy field

The following questions have not always been agreed upon in the super-resolution/SIM field, and with this special issue, we hope to address some of these:

1. What is SRM and should diffraction-limited linear SIM be classified as 'super-resolution'?
2. Should High-NA TIRF-SIM, which can achieve a lateral resolution of down to 84 nm, be considered as diffraction limited?
3. Can nonlinear SIM become broadly applicable and live-cell compatible?
4. Do you need 'switching' of states for nonlinear super-resolution?
5. How can information about single-molecule detection be best combined with the knowledge of the illumination structure?
6. Do high-quality SIM images require reconstruction in Fourier space?
7. Can SIM be used for deep tissue imaging?
8. How can the fundamental limitation of SIM, i.e. generating sufficient stripe contrast in densely labelled and/or extended biological structures due to out-of-focus light, be addressed?
9. Should image scanning microscopy be considered a form of SIM and what forms of structured illumination could be used other than stripes?

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10. Can SIM be used to improve the resolution of (Rayleigh scattering) transmission microscopy?
11. How does sparse illumination compare to dense illumination in linear and nonlinear SIM?
12. Can we generate 'true' super-resolution images from simple instruments enhanced with machine-learning-based algorithms?
13. Can research-grade super-resolution (SIM) microscopes be built cost-efficiently?

For answers to these fundamental questions in super-resolution microscopy, please refer to [85].

3. The wider-reaching social implications

SRM has evolved into a highly interdisciplinary field requiring experts from physics, engineering, chemistry, biology and computer sciences driving innovations to further increase spatial and temporal resolution and ultimately biological applicability. The global advances in SRM need to be synchronized to inspire researchers in the field of microscopy to combine their methods with techniques from different areas.

SRM and SIM, in particular, have become important tools of basic science discovery. However, there are still considerable activation barriers for researchers to venture into SRM, as instruments are not yet 'turn-key', and due to the relatively steep learning curve required to use SIM (or any other SRM method) to its full potential. Furthermore, the application of SRM to study human pathologies and for medical diagnosis is still in its infancy, but will foreseeably play a significant role in the future. New SIM modalities, implementation of AO and machine-learning algorithms will help to democratize SIM and increase its usage in biomedical research.

Super-resolution microscopy is very often an expensive undertaking, thus limiting its broader application. In this meeting, we plan to kick-off the idea of an open-source 'openSIM' system inspired by its derivatives in the light-sheet community. Open discussions where ideas and requirements for a prototype are gathered are a first and significant step to make science not only affordable but available. This special issue shows current progress in SIM development and application.

Data accessibility. This article has no additional data.

Competing interests. We declare no competing interests.

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References

1. Abbe E. 1873 Beitrage zur Theorie des Mikroskops und der mikroskopischen Wahrnehmung. *Arch. Mikr. Anat.* **9**, 413–468. (doi:10.1007/BF02956173)
2. Schermelleh L, Heintzmann R, Leonhardt H. 2010 A guide to super-resolution fluorescence microscopy. *J. Cell Biol.* **190**, 165–175. (doi:10.1083/jcb.201002018)
3. Heintzmann R, Huser T. 2017 Super-resolution structured illumination microscopy. *Chem. Rev.* **117**, 13 890–13 908. (doi:10.1021/acs.chemrev.7b00218)
4. Sheppard CJ. 2017 Resolution and super-resolution. *Microsc. Res. Tech.* **80**, 590–598. (doi:10.1002/jemt.22834)
5. Schermelleh L, Ferrand A, Huser T, Eggeling C, Sauer M, Biehlmaier O, Drummen GP. 2019 Super-resolution microscopy demystified. *Nat. Cell Biol.* **21**, 72–84. (doi:10.1038/s41556-018-0251-8)

Downloaded from https://royalsocietypublishing.org/ on 14 December 2022

6. Lidke KA, Rieger B, Jovin TM, Heintzmann R. 2005 Superresolution by localization of quantum dots using blinking statistics. *Opt. Express* **13**, 7052–7062. (doi:10.1364/OPEX.13.007052)
7. Betzig E, Patterson GH, Sougrat R, Lindwasser OW, Olenych S, Bonifacino JS, Davidson MW, Lippincott-Schwartz J, Hess HF. 2006 Imaging intracellular fluorescent proteins at nanometer resolution. *Science* **313**, 1642–1645. (doi:10.1126/science.1127344)
8. Hess ST, Girirajan TP, Mason MD. 2006 Ultra-high resolution imaging by fluorescence photoactivation localization microscopy. *Biophys. J.* **91**, 4258–4272. (doi:10.1529/biophysj.106.091116)
9. Rust MJ, Bates M, Zhuang X. 2006 Stochastic optical reconstruction microscopy (storm) provides sub-diffraction-limit image resolution. *Nat. Methods* **3**, 793–796. (doi:10.1038/nmeth929)
10. Okhonin V. 1991 A method of examination of sample microstructure. SU 1374922 A1. Priority 10.04.1986. (doi:10.13140/2.1.2588.1922)
11. Hell SW, Wichmann J. 1994 Breaking the diffraction resolution limit by stimulated emission: stimulated-emission-depletion fluorescence microscopy. *Opt. Lett.* **19**, 780–782. (doi:10.1364/OL.19.000780)
12. Baer SC. 1999 Method and apparatus for improving resolution in scanned optical system. US Patent 5 866 911.
13. Heintzmann R, Cremer CG. 1999 Laterally modulated excitation microscopy: improvement of resolution by using a diffraction grating. In *Optical biopsies and microscopic techniques III*, vol. 3568, pp. 185–196. Bellingham, WA: SPIE.
14. Gustafsson MG. 2000 Surpassing the lateral resolution limit by a factor of two using structured illumination microscopy. *J. Microsc.* **198**, 82–87. (doi:10.1046/j.1365-2818.2000.00710.x)
15. Betzig E, Hell SW, Moerner WE. 2014 The nobel prize in chemistry 2014. *Nobel Media AB*.
16. Balzarotti F, Eilers Y, Gwosch KC, Gynnå AH, Westphal V, Stefani FD, Elf J, Hell SW. 2017 Nanometer resolution imaging and tracking of fluorescent molecules with minimal photon fluxes. *Science* **355**, 606–612. (doi:10.1126/science.aak9913)
17. Cnossen J, Hinsdale T, Thorsen RØ, Siemons M, Schueder F, Jungmann R, Smith CS, Rieger B, Stallinga S. 2020 Localization microscopy at doubled precision with patterned illumination. *Nat. Methods* **17**, 59–63. (doi:10.1038/s41592-019-0657-7)
18. Reymond L, Huser T, Ruprecht V, Wieser S. 2020 Modulation-enhanced localization microscopy. *J. Phys.: Photonics* **2**, 041001. (doi:10.1088/2515-7647/ab9eac)
19. Prakash K. 2021 At the molecular resolution with MINFLUX? Preprints. (doi:10.20944/preprints202102.0173)
20. Heintzmann R, Gustafsson MG. 2009 Subdiffraction resolution in continuous samples. *Nat. Photonics* **3**, 362–364. (doi:10.1038/nphoton.2009.102)
21. Schermelleh L *et al.* 2008 Subdiffraction multicolor imaging of the nuclear periphery with 3d structured illumination microscopy. *Science* **320**, 1332–1336. (doi:10.1126/science.1156947)
22. Smeets D *et al.* 2014 Three-dimensional super-resolution microscopy of the inactive x chromosome territory reveals a collapse of its active nuclear compartment harboring distinct xist RNA foci. *Epigenetics Chromatin* **7**, 1–27. (doi:10.1186/1756-8935-7-8)
23. Shaw M, Bella A, Ryadnov MG. 2017 Creim: Coffee ring effect imaging model for monitoring protein self-assembly in situ. *J. Phys. Chem. Lett.* **8**, 4846–4851. (doi:10.1021/acs.jpclett.7b02147)
24. Kner P, Chhun BB, Griffis ER, Winoto L, Gustafsson MG. 2009 Super-resolution video microscopy of live cells by structured illumination. *Nat. Methods* **6**, 339–342. (doi:10.1038/nmeth.1324)
25. Hirvonen LM, Wicker K, Mandula O, Heintzmann R. 2009 Structured illumination microscopy of a living cell. *Eur. Biophys. J.* **38**, 807–812. (doi:10.1007/s00249-009-0501-6)
26. Shao L, Kner P, Rego EH, Gustafsson MG. 2011 Super-resolution 3d microscopy of live whole cells using structured illumination. *Nat. Methods* **8**, 1044–1046. (doi:10.1038/nmeth.1734)
27. Mudry E, Belkebir K, Girard J, Savatier J, Le Moal E, Nicoletti C, Allain M, Sentenac A. 2012 Structured illumination microscopy using unknown speckle patterns. *Nat. Photonics* **6**, 312–315. (doi:10.1038/nphoton.2012.83)
28. Huang X *et al.* 2018 Fast, long-term, super-resolution imaging with hessian structured illumination microscopy. *Nat. Biotechnol.* **36**, 451–459. (doi:10.1038/nbt.4115)

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29. Wu Y, Shroff H. 2018 Faster, sharper, and deeper: structured illumination microscopy for biological imaging. *Nat. Methods* **15**, 1011–1019. (doi:10.1038/s41592-018-0211-z)
30. York AG, Parekh SH, Dalle Nogare D, Fischer RS, Temprine K, Mione M, Chitnis AB, Combs CA, Shroff H. 2012 Resolution doubling in live, multicellular organisms via multifocal structured illumination microscopy. *Nat. Methods* **9**, 749–754. (doi:10.1038/nmeth.2025)
31. York AG, Chandris P, Dalle Nogare D, Head J, Wawrzusin P, Fischer RS, Chitnis A, Shroff H. 2013 Instant super-resolution imaging in live cells and embryos via analog image processing. *Nat. Methods* **10**, 1122–1126. (doi:10.1038/nmeth.2687)
32. Mertz J. 2011 Optical sectioning microscopy with planar or structured illumination. *Nat. Methods* **8**, 811–819. (doi:10.1038/nmeth.1709)
33. Gray RD, Albrecht D, Beerli C, Huttunen M, Cohen GH, White IJ, Burden JJ, Henriques R, Mercer J. 2019 Nanoscale polarization of the entry fusion complex of vaccinia virus drives efficient fusion. *Nat. Microbiol.* **4**, 1636–1644. (doi:10.1038/s41564-019-0488-4)
34. Nozumi M, Nakatsu F, Katoh K, Igarashi M. 2017 Coordinated movement of vesicles and actin bundles during nerve growth revealed by superresolution microscopy. *Cell Rep.* **18**, 2203–2216. (doi:10.1016/j.celrep.2017.02.008)
35. Jacquemet G, Stubb A, Saup R, Miihkinen M, Kremneva E, Hamidi H, Ivaska J. 2019 Filopodome mapping identifies p130cas as a mechanosensitive regulator of filopodia stability. *Curr. Biol.* **29**, 202–216. (doi:10.1016/j.cub.2018.11.053)
36. Brown AC *et al.* 2011 Remodelling of cortical actin where lytic granules dock at natural killer cell immune synapses revealed by super-resolution microscopy. *PLoS Biol.* **9**, e1001152. (doi:10.1371/journal.pbio.1001152)
37. McArthur K *et al.* 2018 BAK/BAX macropores facilitate mitochondrial herniation and mtDNA efflux during apoptosis. *Science* **359**, eaao6047. (doi:10.1126/science.aao6047)
38. Eswaramoorthy P, Winter PW, Wawrzusin P, York AG, Shroff H, Ramamurthi KS. 2014 Asymmetric division and differential gene expression during a bacterial developmental program requires DivIVA. *PLoS Genet.* **10**, e1004526. (doi:10.1371/journal.pgen.1004526)
39. Miron E *et al.* 2020 Chromatin arranges in chains of mesoscale domains with nanoscale functional topography independent of cohesin. *Sci. Adv.* **6**, eaba8811. (doi:10.1126/sciadv.aba8811)
40. Murugesan S *et al.* 2016 Formin-generated actomyosin arcs propel t cell receptor microcluster movement at the immune synapse. *J. Cell Biol.* **215**, 383–399. (doi:10.1083/jcb.201603080)
41. Hübner B, Lomiento M, Mammoli F, Illner D, Markaki Y, Ferrari S, Cremer M, Cremer T. 2015 Remodeling of nuclear landscapes during human myelopoietic cell differentiation maintains co-aligned active and inactive nuclear compartments. *Epigenetics Chromatin* **8**, 1–21. (doi:10.1186/1756-8935-8-1)
42. Sir J-H *et al.* 2011 A primary microcephaly protein complex forms a ring around parental centrioles. *Nat. Genet.* **43**, 1147–1153. (doi:10.1038/ng.971)
43. Manor U, Bartholomew S, Golani G, Christenson E, Kozlov M, Higgs H, Spudich J, Lippincott-Schwartz J. 2015 A mitochondria-anchored isoform of the actin-nucleating spire protein regulates mitochondrial division. *Elife* **4**, e08828. (doi:10.7554/eLife.08828)
44. Guo Y *et al.* 2018 Visualizing intracellular organelle and cytoskeletal interactions at nanoscale resolution on millisecond timescales. *Cell* **175**, 1430–1442. (doi:10.1016/j.cell.2018.09.057)
45. Lesterlin C, Ball G, Schermelleh L, Sherratt DJ. 2014 Reca bundles mediate homology pairing between distant sisters during dna break repair. *Nature* **506**, 249–253. (doi:10.1038/nature12868)
46. Ochs F *et al.* 2019 Stabilization of chromatin topology safeguards genome integrity. *Nature* **574**, 571–574. (doi:10.1038/s41586-019-1659-4)
47. Sonnen KF, Schermelleh L, Leonhardt H, Nigg EA. 2012 3d-structured illumination microscopy provides novel insight into architecture of human centrosomes. *Biol. Open.* **1**, 965–976. (doi:10.1242/bio.20122337)
48. Mennella V, Keszthelyi B, McDonald K, Chhun B, Kan F, Rogers GC, Huang B, Agard D. 2012 Subdiffraction-resolution fluorescence microscopy reveals a domain of the centrosome critical for pericentriolar material organization. *Nat. Cell Biol.* **14**, 1159–1168. (doi:10.1038/ncb2597)

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49. Demmerle J *et al.* 2017 Strategic and practical guidelines for successful structured illumination microscopy. *Nat. Protoc.* **12**, 988–1010. (doi:10.1038/nprot.2017.019)
50. Heintzmann R, Jovin TM, Cremer C. 2002 Saturated patterned excitation microscopy—a concept for optical resolution improvement. *JOSA A* **19**, 1599–1609. (doi:10.1364/JOSAA.19.001599)
51. Gustafsson MG. 2005 Nonlinear structured-illumination microscopy: wide-field fluorescence imaging with theoretically unlimited resolution. *Proc. Natl Acad. Sci. USA* **102**, 13 081–13 086. (doi:10.1073/pnas.0406877102)
52. Rego EH, Shao L, Macklin JJ, Winoto L, Johansson GA, Kamps-Hughes N, Davidson MW, Gustafsson MG. 2012 Nonlinear structured-illumination microscopy with a photoswitchable protein reveals cellular structures at 50-nm resolution. *Proc. Natl Acad. Sci. USA* **109**, E135–E143. (doi:10.1073/pnas.1107547108)
53. Li D *et al.* 2015 Extended-resolution structured illumination imaging of endocytic and cytoskeletal dynamics. *Science* **349**, aab3500. (doi:10.1126/science.aab3500)
54. Reymond L, Ziegler J, Knapp C, Wang F-C, Huser T, Ruprecht V, Wieser S. 2019 Simple: structured illumination based point localization estimator with enhanced precision. *Opt. Express* **27**, 24 578–24 590. (doi:10.1364/OE.27.024578)
55. Gu L, Li Y, Zhang S, Xue Y, Li W, Li D, Xu T, Ji W. 2019 Molecular resolution imaging by repetitive optical selective exposure. *Nat. Methods* **16**, 1114–1118. (doi:10.1038/s41592-019-0544-2)
56. Jouchet P, Cabriel C, Bourg N, Bardou M, Poüs C, Fort E, Lévêque-Fort S. 2019 Nanometric axial localization of single fluorescent molecules with modulated excitation. *BioRxiv*, p. 865865.
57. Hajj B, Wisniewski J, El Beheiry M, Chen J, Revyakin A, Wu C, Dahan M. 2014 Whole-cell, multicolor superresolution imaging using volumetric multifocus microscopy. *Proc. Natl Acad. Sci. USA* **111**, 17 480–17 485. (doi:10.1073/pnas.1412396111)
58. Abrahamsson S *et al.* 2017 Multifocus structured illumination microscopy for fast volumetric super-resolution imaging. *Biomed. Opt. Express* **8**, 4135–4140. (doi:10.1364/BOE.8.004135)
59. Ingaramo M, York AG, Wawrzusin P, Milberg O, Hong A, Weigert R, Shroff H, Patterson GH. 2014 Two-photon excitation improves multifocal structured illumination microscopy in thick scattering tissue. *Proc. Natl Acad. Sci. USA* **111**, 5254–5259. (doi:10.1073/pnas.1314447111)
60. Sheppard CJ, Castello M, Tortarolo G, Vicidomini G, Diaspro A. 2017 Image formation in image scanning microscopy, including the case of two-photon excitation. *JOSA A* **34**, 1339–1350. (doi:10.1364/JOSAA.34.001339)
61. Liu F, Li Q, Jiang S, Zhou L, Zhang J, Zhang H. 2021 Two-photon structured illumination microscopy imaging using fourier ptychography scheme. *Opt. Commun.* **489**, 126872. (doi:10.1016/j.optcom.2021.126872)
62. Keller PJ, Schmidt AD, Santella A, Khairy K, Bao Z, Wittbrodt J, Stelzer EH. 2010 Fast, high-contrast imaging of animal development with scanned light sheet-based structured-illumination microscopy. *Nat. Methods* **7**, 637–642. (doi:10.1038/nmeth.1476)
63. Chang B-J, Meza VDP, Stelzer EH. 2017 csiLSFM combines light-sheet fluorescence microscopy and coherent structured illumination for a lateral resolution below 100 nm. *Proc. Natl Acad. Sci. USA* **114**, 4869–4874. (doi:10.1073/pnas.1609278114)
64. Helle ØI, Dullo FT, Lahrberg M, Tinguely J-C, Hellesø OG, Ahluwalia BS. 2020 Structured illumination microscopy using a photonic chip. *Nat. Photonics* **14**, 431–438. (doi:10.1038/s41566-020-0620-2)
65. Wang H, Lachmann R, Marsikova B, Heinzmann R, Diederich B. 2021 uCsim2: 2D structured illumination microscopy using UC2. *bioRxiv*. (doi:10.1101/2021.01.08.425840)
66. Lachetta M, Sandmeyer H, Sandmeyer A, Esch JS, Huser T, Müller M. 2021 Simulating digital micromirror devices for patterning coherent excitation light in structured illumination microscopy. *Phil. Trans. R. Soc. A* **379**, 20200147. (doi:10.1098/rsta.2020.0147)
67. Schaefer L, Schuster D, Schaffer J. 2004 Structured illumination microscopy: artefact analysis and reduction utilizing a parameter optimization approach. *J. Microsc.* **216**, 165–174. (doi:10.1111/j.0022-2720.2004.01411.x)
68. Ball G, Demmerle J, Kaufmann R, Davis I, Dobbie IM, Schermelleh L. 2015 Simcheck: a toolbox for successful super-resolution structured illumination microscopy. *Sci. Rep.* **5**, 1–12. (doi:10.1038/srep15915)

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69. Mo Y, Feng F, Mao H, Fan J, Chen L. 2021 Structured illumination microscopy artefacts caused by illumination scattering. *Phil. Trans. R. Soc. A* **379**, 20200153. (doi:10.1098/rsta.2020.0153)
70. Yeh L-H, Chowdhury S, Waller L. 2019 Computational structured illumination for high-content fluorescence and phase microscopy. *Biomed. Opt. express* **10**, 1978–1998. (doi:10.1364/BOE.10.001978)
71. Boland MA, Cohen EAK, Flaxman SR, Neil MAA. 2021 Improving axial resolution in Structured Illumination Microscopy using deep learning. *Phil. Trans. R. Soc. A* **379**, 20200298. (doi:10.1098/rsta.2020.0298)
72. Débarre D, Botcherby EJ, Booth MJ, Wilson T. 2008 Adaptive optics for structured illumination microscopy. *Opt. Express* **16**, 9290–9305. (doi:10.1364/OE.16.009290)
73. Booth M, Andrade D, Burke D, Patton B, Zurauskas M. 2015 Aberrations and adaptive optics in super-resolution microscopy. *Microscopy* **64**, 251–261. (doi:10.1093/jmicro/dfv033)
74. Ji N. 2017 Adaptive optical fluorescence microscopy. *Nat. Methods* **14**, 374–380. (doi:10.1038/nmeth.4218)
75. Phillips MA *et al.* 2020 Cryosim: super-resolution 3d structured illumination cryogenic fluorescence microscopy for correlated ultrastructural imaging. *Optica* **7**, 802–812. (doi:10.1364/OPTICA.393203)
76. Müller M, Mönkemöller V, Hennig S, Hübner W, Huser T. 2016 Open-source image reconstruction of super-resolution structured illumination microscopy data in imagej. *Nat. Commun.* **7**, 1–6. (doi:10.1038/ncomms10980)
77. Chen B-C *et al.* 2014 Lattice light-sheet microscopy: imaging molecules to embryos at high spatiotemporal resolution. *Science* **346**, 1257998. (doi:10.1126/science.1257998)
78. Liu T-L *et al.* 2018 Observing the cell in its native state: imaging subcellular dynamics in multicellular organisms. *Science* **360**, eaaq1392. (doi:10.1126/science.aaq1392)
79. Shaw M, Zajiczek L, O'Holleran K. 2015 High speed structured illumination microscopy in optically thick samples. *Methods* **88**, 11–19. (doi:10.1016/j.ymeth.2015.03.020)
80. Song L, Lu-Walther H-W, Förster R, Jost A, Kielhorn M, Zhou J, Heintzmann R. 2016 Fast structured illumination microscopy using rolling shutter cameras. *Meas. Sci. Technol.* **27**, 055401. (doi:10.1088/0957-0233/27/5/055401)
81. Prakash K. 2020 Laser-free super-resolution microscopy. *bioRxiv*.
82. Russell CT, Shaw M. 2021 mmSIM: An open toolbox for accessible structured illumination microscopy. *Phil. Trans. R. Soc. A* **379**, 20200353. (doi:10.1098/rsta.2020.0353)
83. Jin L, Liu B, Zhao F, Hahn S, Dong B, Song R, Elston TC, Xu Y, Hahn KM. 2020 Deep learning enables structured illumination microscopy with low light levels and enhanced speed. *Nat. Commun.* **11**, 1–7. (doi:10.1101/866822)
84. Ling C, Zhang C, Wang M, Meng F, Du L, Yuan X. 2020 Fast structured illumination microscopy via deep learning. *Photonics Res.* **8**, 1350–1359. (doi:10.1364/PRJ.396122)
85. Heintzmann R. 2021 Answers to fundamental questions in superresolution microscopy. *Phil. Trans. R. Soc. A* **379**, 20210105. (doi:10.1098/rsta.2021.0105)